

Rafi Khan

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Education

The Ohio State University

Master of Science in Mechanical Engineering, GPA: 4.0/4.0

Expected December 2026

Bachelor of Science in Mechanical Engineering, GPA: 3.68/4.0

December 2024

Experience

Materials at Extremes Lab, OSU

Jan 2025 – Present

Graduate Research Assistant

- Engineered a novel predictive algorithm to accelerate material qualification by forecasting long-term creep behavior from partial experimental data, reducing characterization time for Inconel Superalloys by ~67%
- Implemented the model in MATLAB, leveraging the Optimization Toolbox's non-linear solvers and the Ohio Supercomputer (HPC) to accurately predict 7500-hour material life from only 2500 hours of test data. (Results in preparation for ASME Turbo Expo 2026)
- Designed and validated a novel multi-stress creep specimen using ANSYS FEA to generate 3 distinct, uniform stress regions, increasing the experimental data throughput by 3x compared to standard methods (ASTM E139)

SpaceX, Hawthorne, CA

May 2023 – August 2023

Supplier Development Engineering Intern, Starlink Aviation

- Implemented high voltage testing of Starlink harnesses for the first time in the department. Owned all the processes from test hardware setup, fabrication, programming, and implementation in the reliability lab.
- Uncovered novel defects when performing tests per Federal Aviation Administration (FAA) requirements
- Led the manufacturing qualification for a new Starlink Aviation product, identifying critical gaps in the supplier's process and correcting their tensile-strength test procedures to achieve IPC-620 compliance
- Identified a non-conformance (NC) on over 3000 in-stock harnesses planned for installation on Airbus A320s, placing NCs into containment and passing FAA certification

General Electric Aerospace, Dayton, OH

September 2022 – December 2022

Manufacturing Engineering Intern, Aviation Systems

- Assembled high-voltage (1500 V) power converters for a new hybrid aircraft program by interpreting complex engineering drawings and design requirements
- Authored detailed work instructions based on the initial build process, creating a repeatable standard that streamlined assembly for technicians on future units
- Designed and fabricated a custom screen-printing jig to apply thermal paste onto sensitive semiconductor modules, establishing a repeatable process to ensure reliable thermal management
- Acquired and applied high-reliability skills by fabricating aerospace-grade wire harnesses, terminating fiber optic cables, and performing certified soldering according to IPC-610 standards

Denso, Maryville, TN

May 2022 – August 2022

Process Design Intern

- Responsible for optimizing a high-volume automated production line of computer modules used in 90% of General Motor vehicles
- Designed a calibration jig in SolidWorks that eliminated a bottleneck in final packaging, reducing machine downtime and saving an average of \$1,000 per day
- Used Cpk analysis to diagnose dimensional flaws in the cardboard supply, resolving automated line crashes with a new box orientation procedure

Skills

SolidWorks (Certified), MATLAB, Ansys, nCode DesignLife, HSMWorks (CAM), LabVIEW, Python, 3D-Printing